

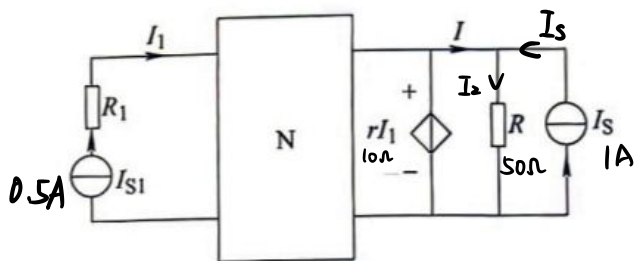
顾峻熹 3250104213 U₁

1.4 1-4 题图1-4所示电路中, $I_{S1} = 0.5A$, $I_S = 1A$, 控制系数 $r = 10\Omega$, 电阻 $R = 50\Omega$ 。方框内为任意电路 (设不短路), 试求电流 I 。

解. $I_1 = I_{S1} = 0.5A$

$$rI_1 = 5V = I_2 R = (I + I_S) R$$

$$I = 0.1A - 1A = -0.9A \quad ①$$



题图 1-4

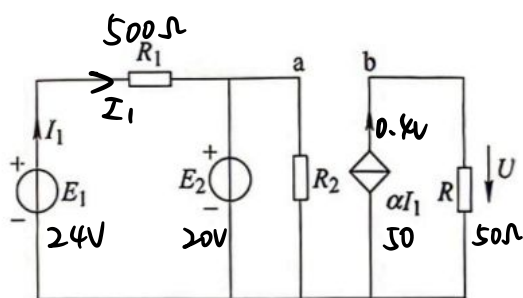
1.6 1-6 题图1-6所示电路中, 电压源 $E_1 = 24V$, $E_2 = 20V$, $\alpha = 50$, $R = 50\Omega$, $R_1 = 500\Omega$, 试求 U_{ab} 和 U 。

解: $E_1 - I_1 R_1 - E_2 = 0$

$$\Rightarrow I_1 = 0.008A$$

$$U = \alpha I_1 R = 20V \quad ②$$

$$U_{ab} = E_2 - U = 0V \quad ①$$



题图 1-6

1.7 1-7 题图1-7所示电路中, $R_1 = R_2 = R_3 = R_4 = 40\Omega$, $R_5 = 20\Omega$, $E_1 = 100V$, $E_2 = 20V$ 。欲使电压源 E_2 中无电流通过, 求 R_X 之值。

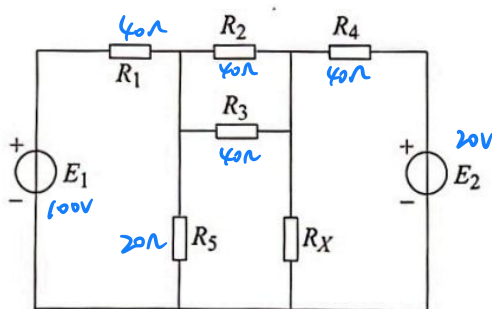
$$R_{23} = \frac{R_2 R_3}{R_2 + R_3} = 20\Omega \quad \text{化简电路图.}$$

$$KCL: \begin{cases} I_1 = I_2 + I_3 \\ I_3 = I_4 + I_5 \end{cases}$$

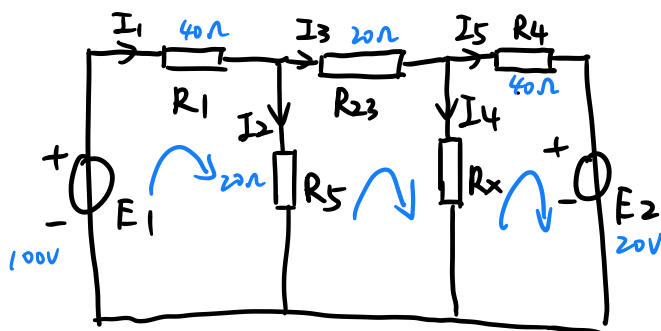
$$KVL: \begin{cases} E_1 - I_1 R_1 - I_2 R_5 = 0 \\ I_2 R_5 - I_3 R_{23} - I_4 R_X = 0 \\ I_4 R_X - I_5 R_4 - E_2 = 0 \end{cases}$$

$$I_5 = 0. \Rightarrow \begin{cases} I_1 = 1.8A \\ I_2 = 1.4A \\ I_3 = 0.4A \\ I_4 = 0.4A \\ R_X = 50\Omega \end{cases}$$

$$\therefore R_X = 50\Omega$$



题图 1-7



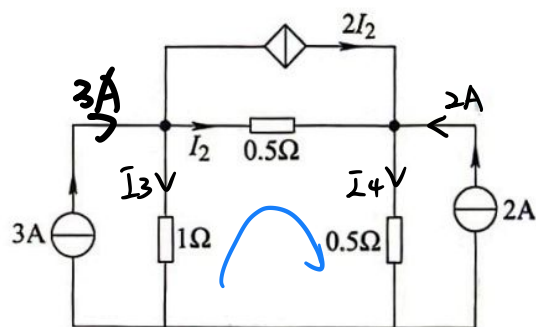
1-8 1-8 电路各参数均标于题图1-8所示电路上，试求 I_2 为多少？

解: KCL:
$$\begin{cases} 3A = I_2 + 2I_2 + I_3 \\ 2A + I_2 + 2I_2 = I_4 \end{cases}$$

KVL:
$$I_3 \cdot 1\Omega - I_2 \cdot 0.5\Omega - I_4 \cdot 0.5\Omega = 0$$

$$\Rightarrow \begin{cases} I_2 = 0.4A \\ I_3 = 1.8A \\ I_4 = 3.2A \end{cases}$$

$$\therefore I_2 = 0.4A$$



题图 1-8

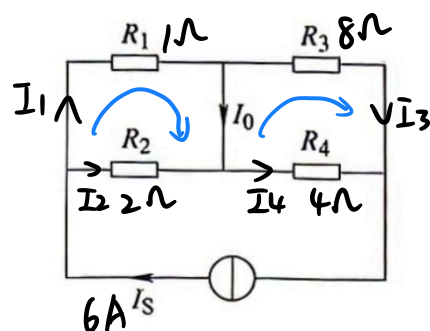
1-11 1-11 题图1-11所示电路中，电流源 $I_s = 6A$ ，电阻 $R_1 = 1\Omega$ ， $R_2 = 2\Omega$ ， $R_3 = 8\Omega$ ， $R_4 = 4\Omega$ ，求电流 I_0 。

解: KCL:
$$\begin{cases} I_s = I_1 + I_2 \\ I_1 = I_0 + I_3 \\ I_2 + I_0 = I_4 \end{cases}$$

KVL:
$$\begin{cases} -I_1 R_1 + I_2 R_2 = 0 \\ -I_3 R_3 + I_4 R_4 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} I_1 = 4A \\ I_2 = 2A \\ I_0 = 2A \\ I_3 = 2A \\ I_4 = 4A \end{cases}$$

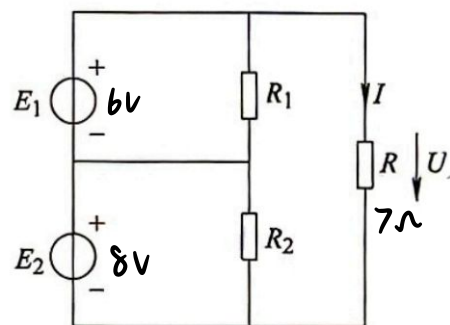
$$\therefore I_0 = 2A$$



题图 1-11

1.16 1-16 题图1-16所示电路中，电压源分别为 $E_1 = 6V$, $E_2 = 8V$, $R = 7\Omega$, 试求电流 I 。

解: $I = \frac{E_1 + E_2}{R} = 2A$



题图 1-16

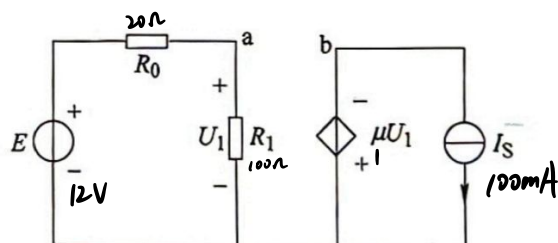
1-18 题图1-18所示电路中，电压源 $E = 12V$, 电流源 $I_S = 100mA$, 电压控制电压源的控制系数 $\mu = 1$, $R_0 = 20\Omega$, $R_1 = 100\Omega$, 试求 U_{ab} 和电流源 I_S 发出的功率。

解: $U_1 = \frac{E}{R_0 + R_1} \cdot R_1 = 10V$

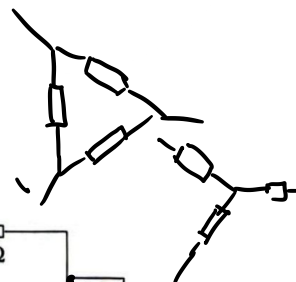
$\mu U_1 = 10V$

$U_{ab} = U_1 + \mu U_1 = 20V$

$P_{I_S} = \mu U_1 \cdot I_S = 1W$



题图 1-18



1-24 题图1-24所示电路为电桥电路，求：

(1) 电压 U 。

(2) 电压 U_{ab} 。

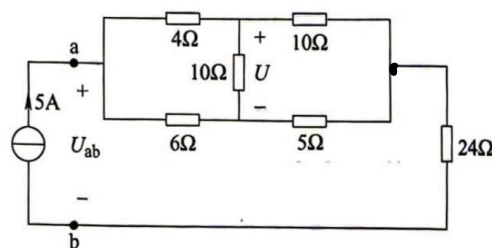
解: 作 Δ -Y 等效变换

$R_1 = \frac{10 \times 10}{10 + 10 + 5} = 4\Omega$

$R_2 = R_3 = \frac{10 \times 5}{10 + 10 + 5} = 2\Omega$

$U_{ab} = 5A \times (\frac{8\Omega \cdot 8\Omega}{8\Omega + 8\Omega} + 2\Omega + 24\Omega) = 150V$ ②

$U = 6\Omega \times 2.5A - 4\Omega \times 2.5A = 5V$ ①



题图 1-24

